

Homework 3

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Problem 1 Suppose that the loss L , due to the arrival of certain type of insurance claims, is distributed according to a Pareto distribution. Specifically, it has been observed that the distribution of L is satisfactorily modeled by the cumulative distribution function,

$$P(L \leq x) = \begin{cases} 1 - 1/x^3, & \text{if } x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

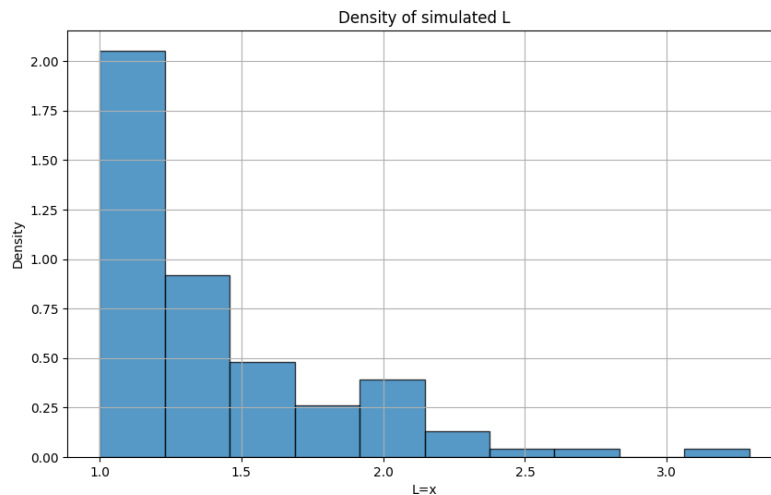
Given access only to a procedure that generates $\text{Unif}(0,1)$, write an algorithm to generate L . Implement the algorithm on a computer to simulate 100 copies of L and plot their distribution.

Solution:

To generate samples of L , we can employ the ITM using the inverse of the CDF of L , which can be derived as follows:

$$\begin{aligned} u &= 1 - \frac{1}{t^3} \\ \Rightarrow t^3 &= \frac{1}{1-u} \\ \Rightarrow t &= \sqrt[3]{\frac{1}{1-u}}. \end{aligned}$$

To generate this random variable L , we proceed as follows: generate $U \sim \text{Unif}(0,1)$, and then apply the inverse transform $L = \sqrt[3]{\frac{1}{1-U}}$.



Problem 2 We consider generating a random variable having density function

$$f(x) = \frac{e^x}{e-1}, 0 < x < 1.$$

- (a) Write the inverse transform method for generating this random variable.
- (b) Write the acceptance-rejection method, with a proposal density $\text{Unif}[0, 1]$.
- (c) Which method in part (a) or (b) do you prefer? Explain your answer.
- (d) Implement the algorithms in both parts (a) and (b) on a computer. Using each algorithm, simulate 100 copies of L and plot their distribution.

Solution:

(a)

For the given pdf, we can write the cdf as follows:

$$F(t) = \int_0^t \frac{e^x}{e-1} dx = \frac{e^t - 1}{e - 1}$$

for $t \in [0, 1]$. $F(t) = 0$ and $F(t) = 1$ for any t such that $t < 0$ and $t > 1$, respectively. We can find the inverse function as follows:

$$u = \frac{e^t - 1}{e - 1}$$

$$\ln(1 + u(e - 1)) = t$$

To generate this random variable, we proceed as follows: generate $U \sim \text{Unif}(0, 1)$, and then apply the inverse transform $X = \ln(1 + U(e - 1))$.

(b)

We first need to find C such that

$$f(x) \leq Cg(x) = C$$

for all $x \in (0, 1)$ where $g(x)$ is the pdf of the uniform distribution. We can simply pick $C = 2$ since

$$f(x) \leq \frac{e}{e-1} < 2.$$

The acceptance-rejection method for generating this random variable is as follows:

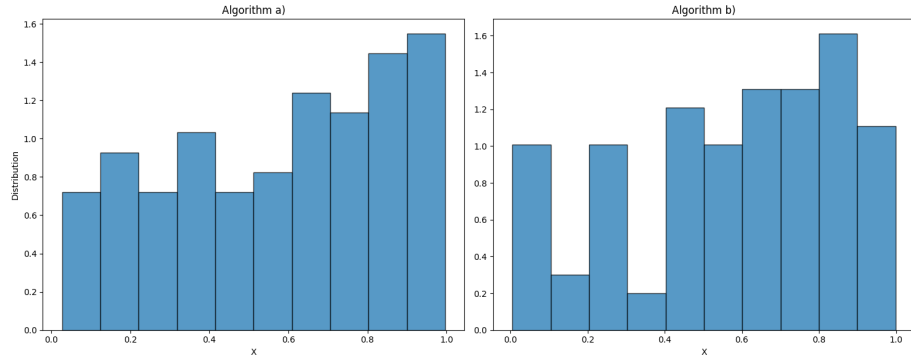
1. Generate $Y \sim \text{Unif}[0, 1]$
2. Generate $U \sim \text{Unif}[0, 1]$
3. If $U \leq \frac{f(Y)}{Cg(Y)} = \frac{e^Y}{2(e-1)}$, then $X = Y$; otherwise, go to the first step
4. Output X .

(c)

One possible answer is that the first method is more preferable as the procedure is much simpler. We can draw a uniform random variable and directly apply the inverse transform method. Moreover, the second method takes multiple random number generations and it might require more than one step if the first round is rejected.

That said, one might argue that the second approach is preferred if they are not comfortable with deriving the cdf and its inverse.

(d) See the plot below



Problem 3 A Gamma distribution with parameters $(n, 1)$ (denoted by $\text{Gamma}(n, 1)$) has density function

$$f(x) = \begin{cases} e^{-x} \frac{x^{n-1}}{(n-1)!}, & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

(a) Derive the acceptance-rejection method to generate a random variable X with distribution $\text{Gamma}(n, 1)$, by using an exponential density function with parameter $1/2$, i.e.,

$$g(x) = \frac{1}{2}e^{-x/2}, \quad x \geq 0.$$

Assume you have access only to a procedure that can generate $\text{Unif}(0, 1)$.

(b) Do you think using an exponential density with parameter 1 as the proposal would work in part (a)? Explain your answer.

Solution:

(a) We first need to find a constant C that upper bound

$$\frac{f(x)}{g(x)} = \frac{e^{-x} \frac{x^{n-1}}{(n-1)!}}{e^{-x/2}/2} = 2e^{-x/2} \frac{x^{n-1}}{(n-1)!}.$$

We notice that this term is bounded as the exponential term will dominate the polynomial term when x gets larger.

To find this constant C , we want to find x^* that maximizes and set $C = \frac{f(x^*)}{g(x^*)}$.

To simplify the differentiation, we use the fact that x^* maximizes the ratio $\frac{f(x)}{g(x)}$ if and only if it maximizes the log of the ratio, i.e., $\log\left(\frac{f(x)}{g(x)}\right)$ by monotonicity of logarithm. Thus, we look at the first-order condition (to be rigorous, you should also check the second-order condition to ensure we are finding a maximum) and see

$$\begin{aligned} 0 &= \frac{d}{dx} \log\left(\frac{f(x)}{g(x)}\right) \Big|_{x=x^*} \\ &= \frac{d}{dx} \left(2 - \frac{x}{2} + (n-1)\log(x) - \log((n-1)!)\right) \Big|_{x=x^*} \\ &= \left(-\frac{1}{2} + \frac{(n-1)}{x^*}\right) \Rightarrow x^* = 2(n-1). \end{aligned}$$

Therefore, we can pick C to be $\frac{f(2(n-1))}{g(2(n-1))}$.

Lastly, we need to generate this exponential random variable with parameter $1/2$ using a uniform random variable. We can do so by directly apply the ITM. The inverse function is $Y = -\frac{\log(1-U)}{\lambda} = 2 \log(1-U)$.

We can implement the acceptance/rejection method for this problem as follows:

1. Set $C = \frac{f(2(n-1))}{g(2(n-1))}$ or some larger value
2. Generate $U_1 \sim Unif[0, 1]$
3. Generate $U_2 \sim Unif[0, 1]$
4. Set $Y = -2 \log(1 - U_1)$
5. If $U_2 \leq \frac{f(Y)}{Cg(Y)}$, then $X = Y$; otherwise, go to the second step
6. Output X .

(b)

No, it would not work. It is impossible to find a finite constant C that satisfies $\frac{f(x)}{g(x)} \leq C$ as we can see from:

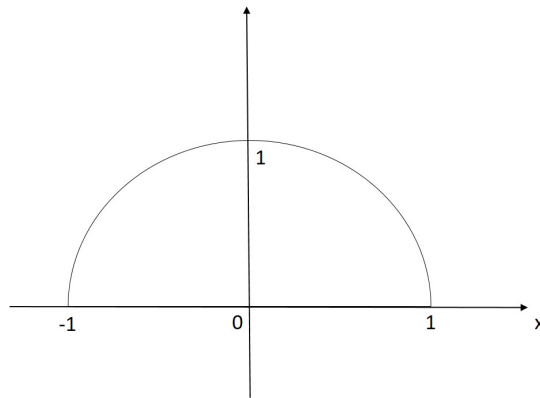
$$\sup_{x \geq 0} \frac{f(x)}{g(x)} = \sup_{x \geq 0} \frac{e^{-x} \frac{x^{n-1}}{(n-1)!}}{e^{-x}} = \sup_{x \geq 0} \frac{x^{n-1}}{(n-1)!} = \infty.$$

Intuitively, the exponential density with parameter 1 has a tail lighter than the target distribution.

Problem 4 The random variable X has a *semi-circular* distribution, with the probability density function (p.d.f) $f(x)$ given by

$$f(x) = \begin{cases} \frac{2}{\pi} \sqrt{1-x^2} & \text{if } -1 \leq x \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

A plot of the p.d.f is given below.



(a) Derive the acceptance-rejection method to generate the random variable X , by using $Unif(-1,1)$ as the proposal. Assume you have access only to a procedure that can generate $Unif(0,1)$.

(b) What is the expected runtime of the method, in terms of the number of acceptance-rejection trials needed to get one output?

Solution:

(a)

First, we need to find a way to generate a sample from $Unif(-1, 1)$. Given that we can generate $X \sim Unif(0, 1)$. We can set $Y = 2X - 1 \sim Unif(-1, 1)$.

Next, we will find a constant C that is greater than the ratio between the semi-circular distribution and the uniform distribution over $[-1, 1]$ by finding

$$\max_{x \in [-1, 1]} \frac{\frac{2}{\pi} \sqrt{1-x^2}}{\frac{1}{2}} = \max_{x \in [-1, 1]} \frac{4}{\pi} \sqrt{1-x^2} = \frac{4}{\pi}.$$

Thus, we can implement the acceptance-rejection as follows:

1. Set $C = \frac{4}{\pi}$
2. Generate $U_1 \sim Unif[0, 1]$
3. Generate $U_2 \sim Unif[0, 1]$
4. Set $Y = 2U_1 - 1$
5. If $U_2 \leq \frac{f(Y)}{\frac{1}{2} \cdot C} = \frac{\frac{2}{\pi} \sqrt{1-Y^2}}{\frac{4}{2\pi}} = \sqrt{1-Y^2}$, then $X = Y$; otherwise, go to the second step
6. Output X .

(b)

As discussed in class, the expected runtime for generating one output is C , which is $\frac{4}{\pi}$.